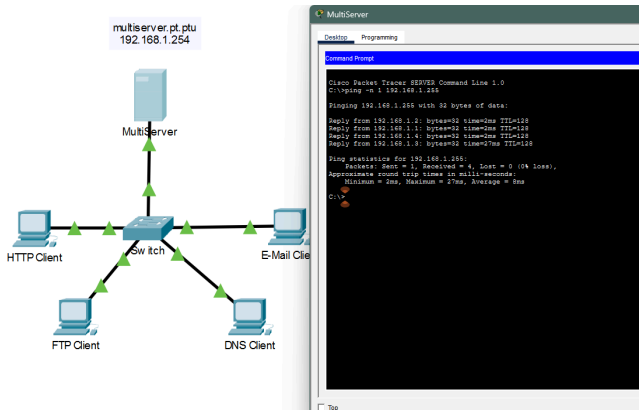
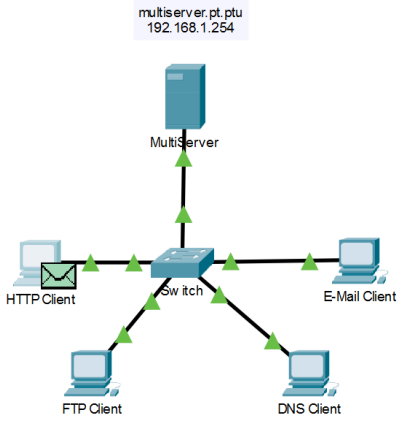
Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

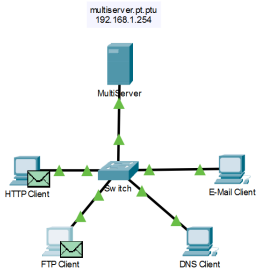
Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.



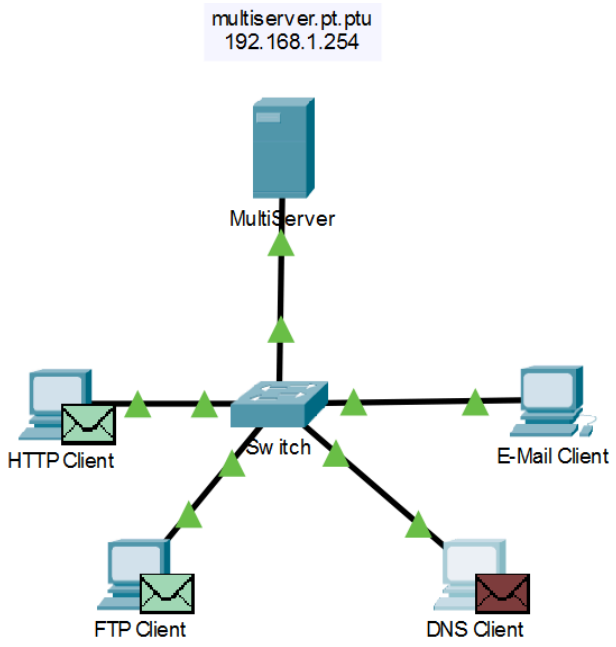
Step 2: Generate web (HTTP) traffic.



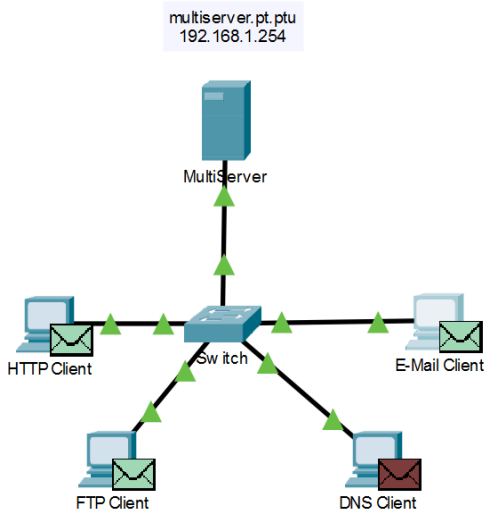
Step 3: Generate FTP traffic.



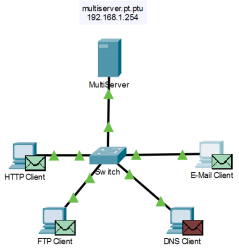
Step 4: Generate DNS traffic.



Step 5: Generate Email traffic.

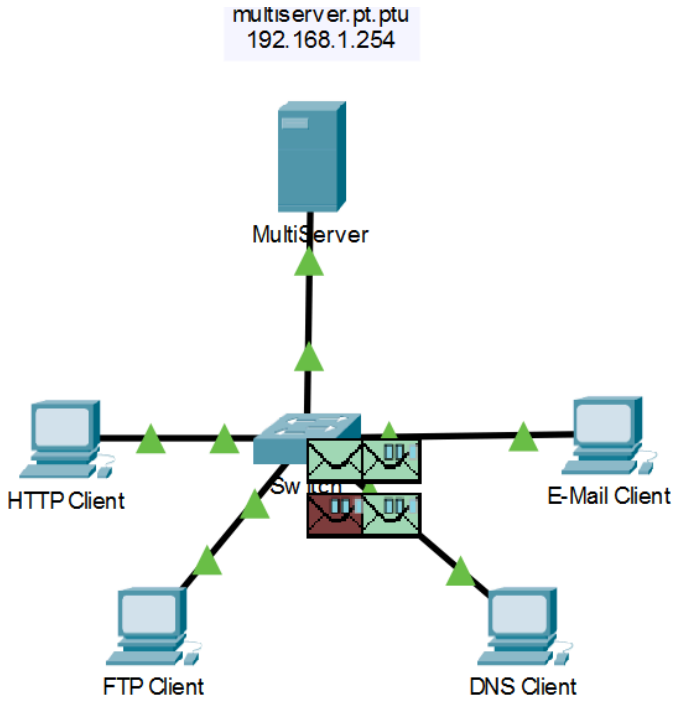


Step 6: Verify that the traffic is generated and ready for simulation.

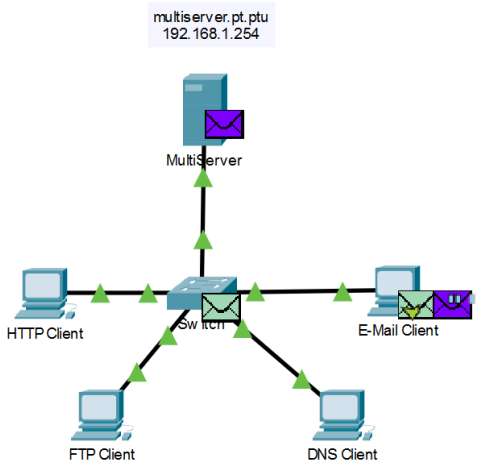


Step 7: Examine multiplexing as the traffic crosses the network.

a. Click Capture/Forward once. All of the PDUs travel to the switch.



b. Click Capture/Forward six times and watch the PDUs from the different hosts as they travel on the network. Note that only one PDU can cross a wire in each direction at any given time.

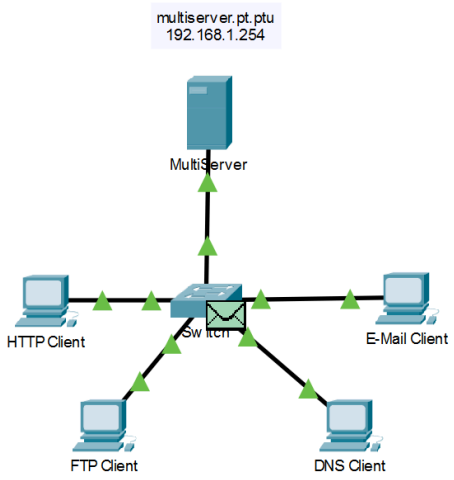


What is this called?
conversation multiplexing.

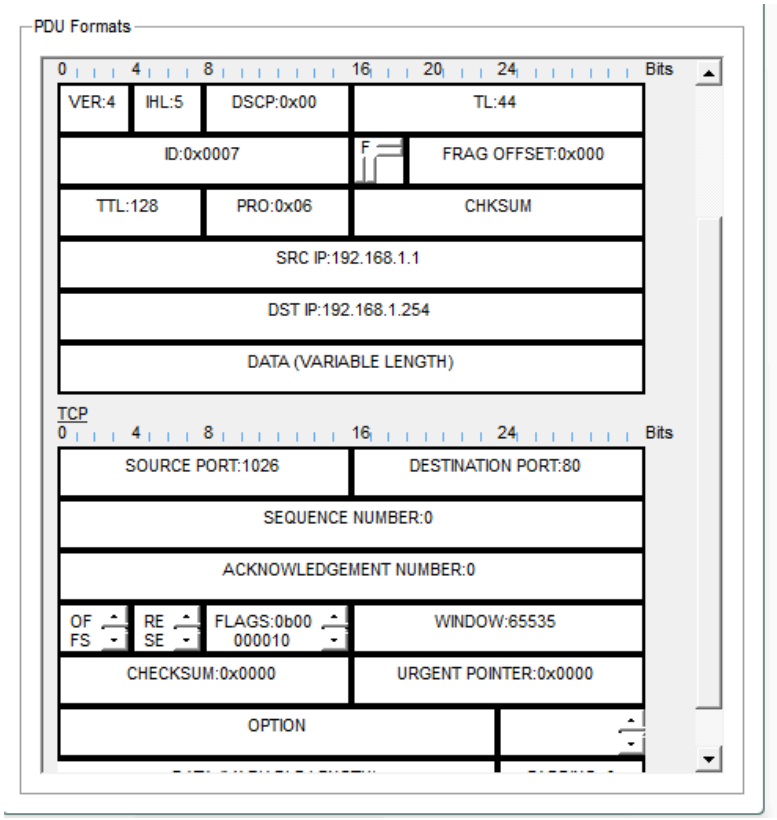
A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?
They represent different protocols.

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine HTTP traffic as the clients communicate with the server.



Why did it take so long for the HTTP PDU to appear?
Because TCP must first establish the connection so that the HTTP traffic can begin.



What is the section labeled?
TCP

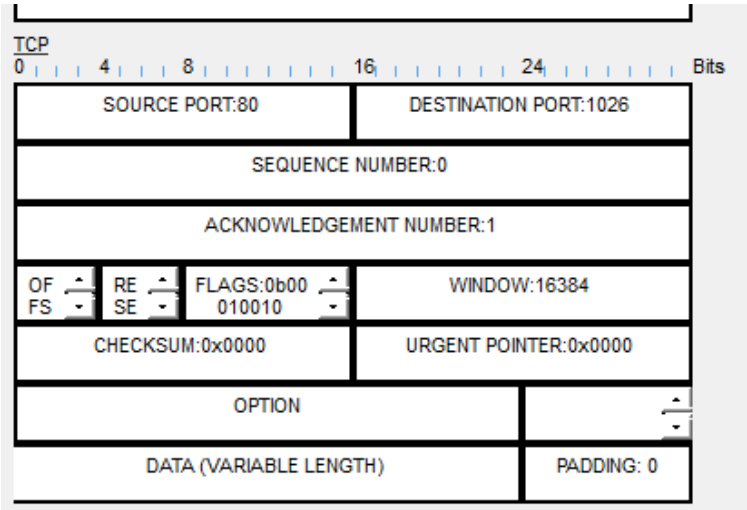
Are these communications considered to be reliable?
Yes, TCP is in use.

Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.
1026 , 80, 0, 0

Which TCP flags are set in this PDU?
ACK and PSH

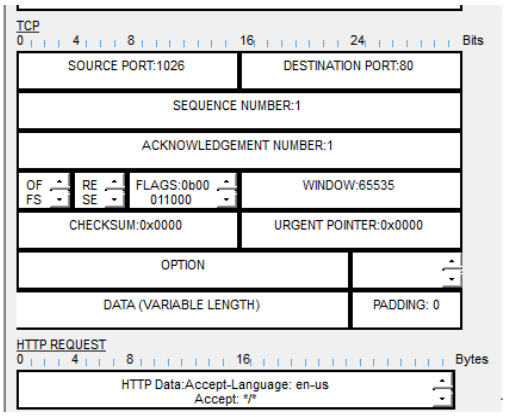
g. Close the PDU and click **Capture/Forward** until a PDU with a checkmark returns to the **HTTP Client**.

h. Click the PDU envelope and select **Inbound PDU Details**.



How are the port and sequence numbers different than before?
The source and destination ports are reversed, and the acknowledgement number is 1. The flags have changed to SYN+ACK.

i. Click the HTTP PDU which HTTP Client has prepared to send to MultiServer. This is the beginning of the HTTP communication. Click this second PDU envelope and select Outbound PDU Details.



What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?
The source and destination ports are reversed, both sequence number is 1, the acknowledgement number is 103 (value my vary), and the flags are PSH and ACK.

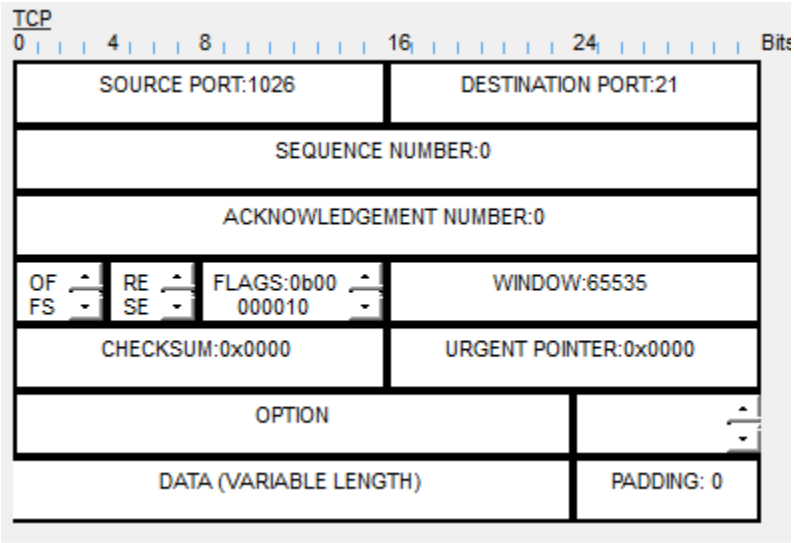
Step 2: Examine FTP traffic as the clients communicate with the server.

a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering ftp 192.168.1.254.

b. In the Simulation Panel, change Edit Filters to display only FTP and TCP.

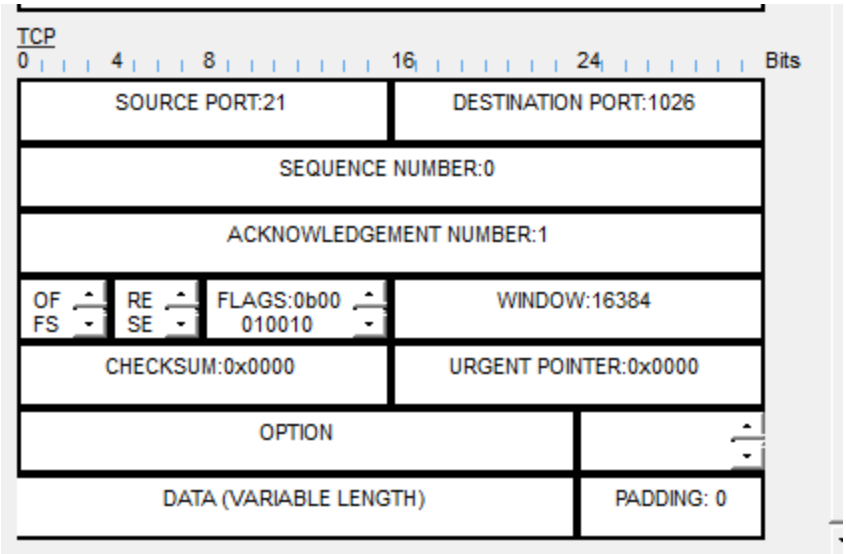
c. Click Capture/Forward. Click the second PDU envelope to open it.
Click the Outbound PDU Details tab and scroll down to the TCP section.

Are these communications considered to be reliable?
Yes



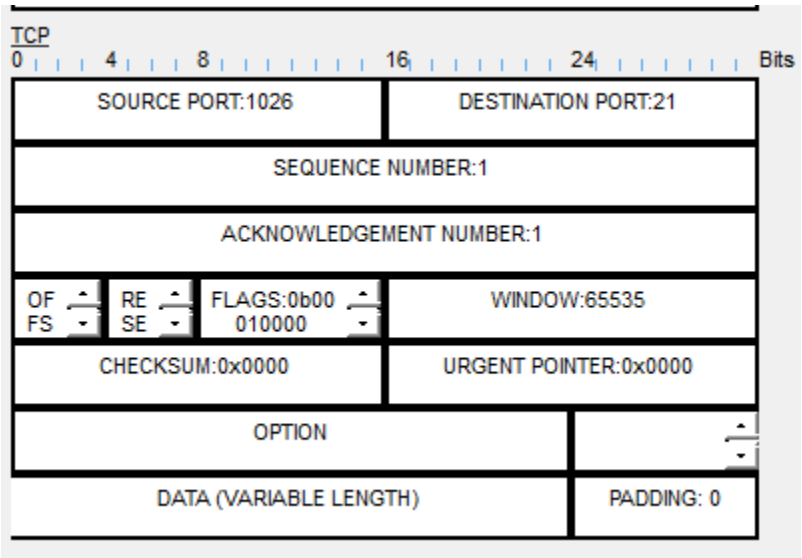
d. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values. What is the flag field value?
1026 , 25, 0, 0. SYN

e. Close the PDU and click Capture/Forward until a PDU returns to the FTP Client with a checkmark.
f.
Click the PDU envelope and select Inbound PDU Details.
Question:
How are the port and sequence numbers different than before?
21, 1026, 0, 1. SYN+ACK. The source and destination ports are reversed, and the acknowledgement number is 1.



h. Click the **Outbound PDU Details** tab.

How are the port and sequence numbers different from the previous two results?
1025, 21, 1, 1. The source and destination ports are reversed, and both sequence and acknowledgement numbers are 1.



h. Close the PDU and click Capture/Forward until a second PDU returns to the FTP Client. The PDU is a different color.

i. Open the PDU and select Inbound PDU Details. Scroll down past the TCP section.

What is the message from the server?
"Welcome to PT Ftp server"

j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to create DNS traffic.

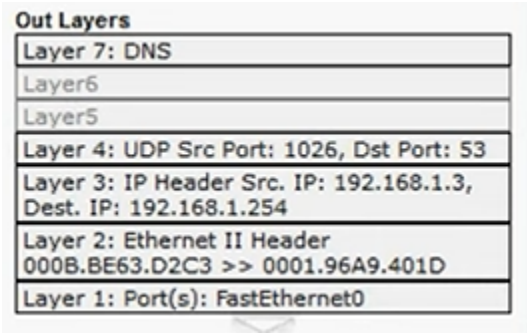
b. In the Simulation Panel, change Edit Filters to display only DNS and UDP.

c. Click the PDU envelope to open it.

d. Look at the OSI Model details for the outbound PDU.

What is the Layer 4 protocol?

UDP



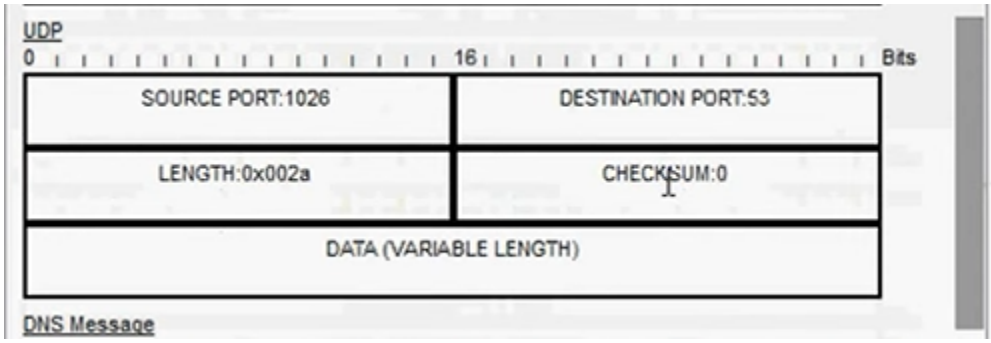
Are these communications considered to be reliable?

No

e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the SRC PORT and DEST PORT values.

Why are there no sequence and acknowledgement numbers?

1026 and 53. Because UDP does not need to establish a reliable connection.

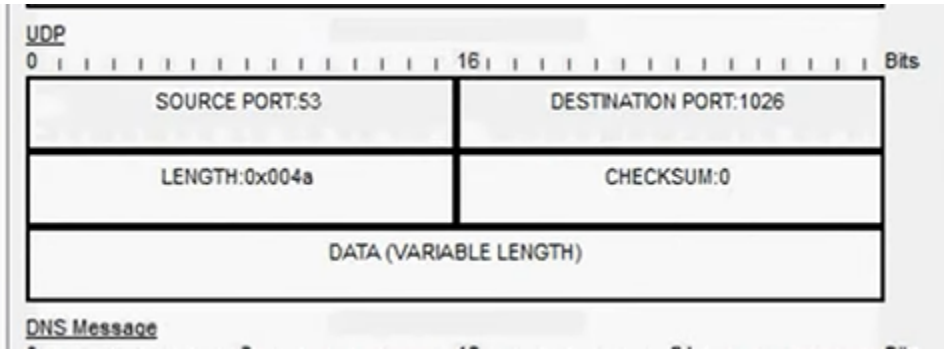


f. Close the PDU and click Capture/Forward until a PDU with a check mark returns to the DNS Client.

g. Click the PDU envelope and select Inbound PDU Details.

How are the port and sequence numbers different than before?

53, 1026. The source and destination ports are reversed.



What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?

DNS ANSWER, 192.1681.254.

h. Click Reset Simulation.

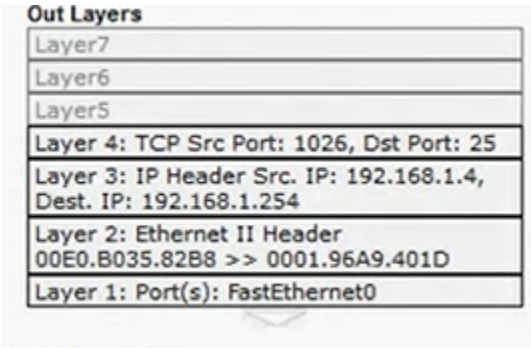
Step 4: Examine email traffic as the clients communicate with the server.
a. Repeat the steps in Part 1 to send an email to user@multiserver.pt.ptu.

b. In the Simulation Panel, change Edit Filters to display only POP3, SMTP and TCP.

c. Click the first PDU envelope to open it.

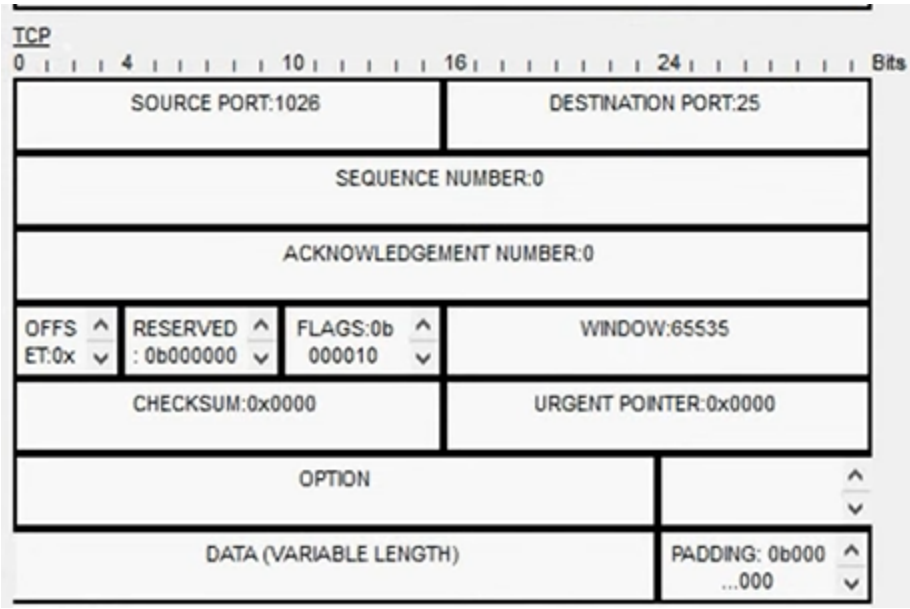
d. Click the Outbound PDU Details tab and scroll down to the last section.

What transport layer protocol does email traffic use?
TCP



Are these communications considered to be reliable?
Yes.

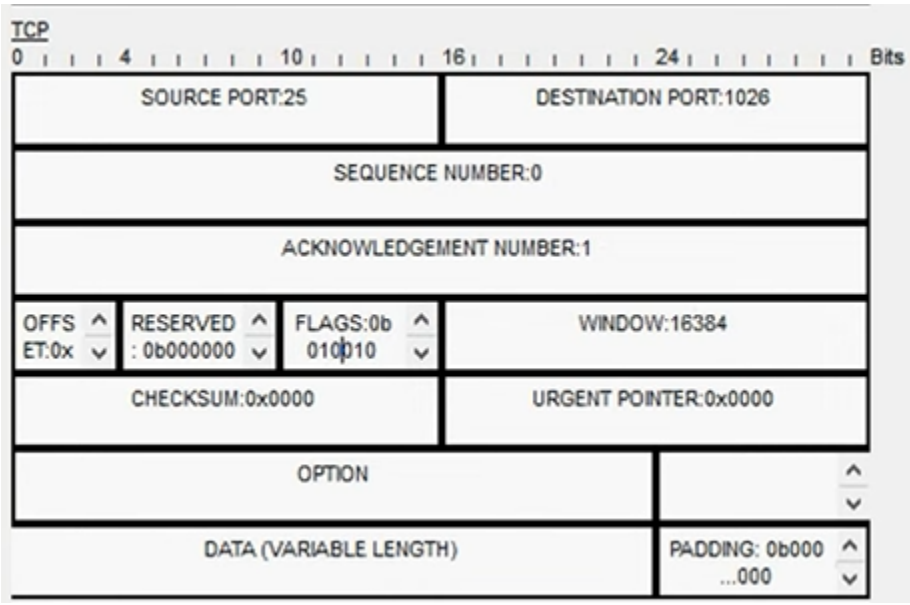
e. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values. What is the flag field value?
1026 , 25, 0, 0. SYN



f. Close the PDU and click Capture/Forward until a PDU returns to the E-Mail Client with a checkmark.

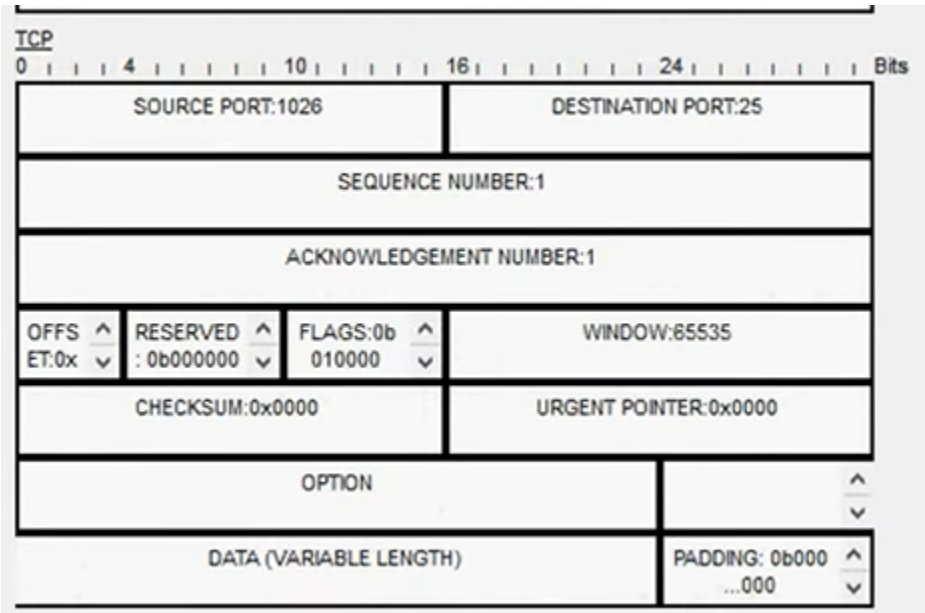
g. Click the TCP PDU envelope and select Inbound PDU Details.

How are the port and sequence numbers different than before?
25, 1026, 0, 1. SYN+ACK. The source and destination ports are reversed, and the acknowledgement number is 1.



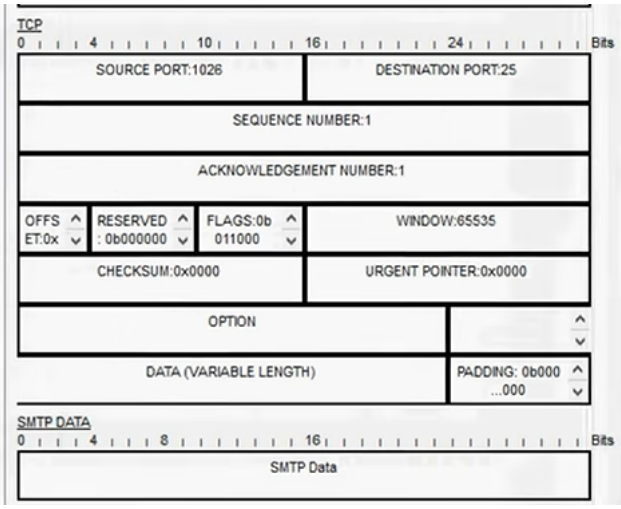
h. Click the Outbound PDU Details tab.

How are the port and sequence numbers different from the previous two results?
1026, 25, 1, 1. ACK. The source and destination ports are reversed, and both sequence and acknowledgement numbers are 1. ACK



i. There is a second PDU of a different color that E-Mail Client has prepared to send to MultiServer. This is the beginning of the email communication. Click this second PDU envelope and select Outbound PDU Details.

How are the port and sequence numbers different from the previous two PDUs?
1026, 25, 1, 1. PSH+ACK. The source and destination ports are reversed, and both sequence and acknowledgement numbers are 1.



What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

SMTP. POP3.